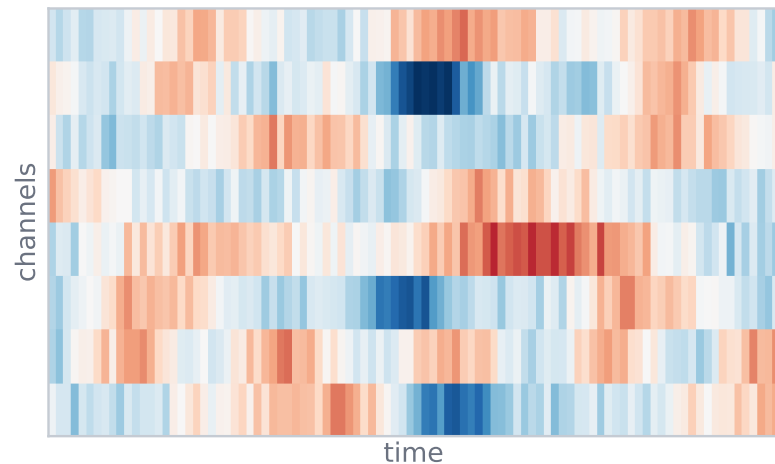


NFPA inductive bias: separable channel-time structure

A EMPCA basis vector p_i

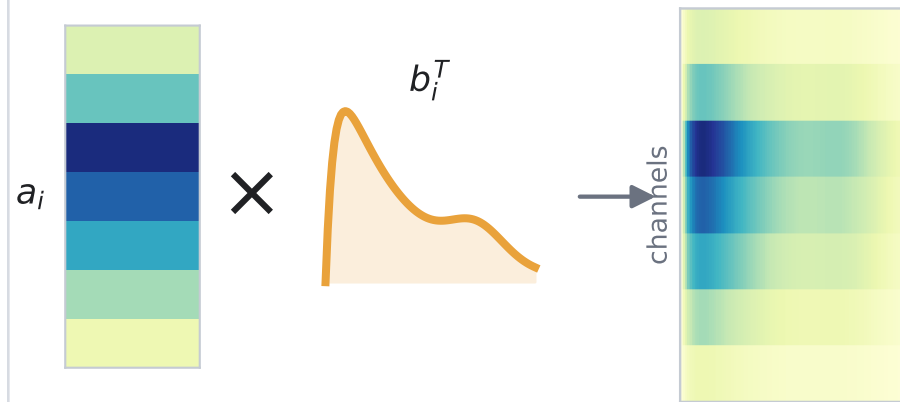
arbitrary channel-time pattern



one free pattern across all channels and all time samples

B NFPA factorization

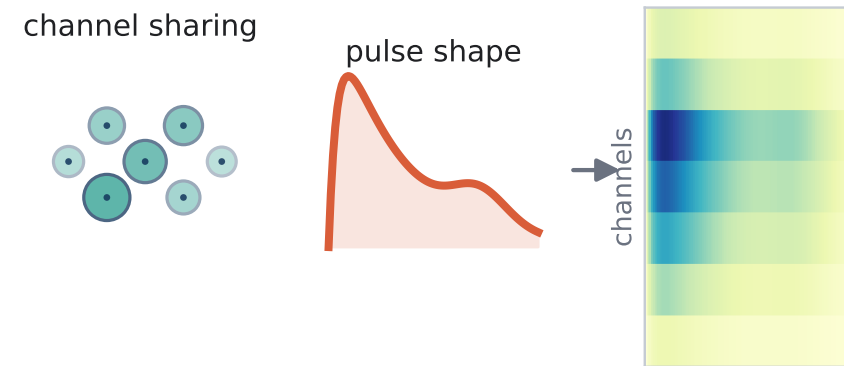
basis constrained to be separable



NFPA basis = channel mode $\times$ time mode

C Detector interpretation

spatial sharing and pulse evolution



NFPA learns spatial sharing $\times$ pulse evolution